13.9 Classifying Quadrilaterals – Part 2

Lesson Introduction

Benchmarks	MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals. MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.	Learning Targets	• I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context. • I can use coordinate geometry to solve geometric problems involving quadrilaterals in a mathematical or real-world context.

Benchmark Coherence	Building On	Working Toward:
	MA.912.AR.2.3 Write a linear, two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.	N/A

Essential Question	• How can you classify a quadrilateral in the real-world?
Lesson Narrative	Lesson 13.9 continues work with quadrilaterals, where students classify a quadrilateral that is in real-world context. Students will classify and justify their classification using properties.

Component Summary				
	13.9.1 Warm-Up (~5 mins)	Between Two Numbers *(SE p. 334)*	13.9.3 Your Turn! (10 mins)	Classifying quadrilaterals independently *(SE p. 338)*
	13.9.2 Collaboration (25 mins)	Real-World Quadrilaterals *(SE p. 335)*	13.9.4 Wrap-Up (~5 mins)	Self-Reflection *(SE p. 339)*

Resources	Glossary	Materials	Additional Preparation
	Key Terms: N/A Additional Terms: • Quadrilateral	• Calculators • (Optional) Graph paper or Graphing Utility	• If wanted, teachers can have the shapes graphed using GeoGebra ahead of the lesson on the board or on the internet, so it can be shown as a visual.

Teacher Tips

Common Misconceptions

- Students may get the properties mixed up according to what quadrilateral they belong to because some overlap.
- Students may miscalculate the length measurements by not leaving them in radical form or rounding as stated.
- Students may round values too early.

Things to Consider

- Consider if, during 13.9.2 Collaboration, students may benefit from you pulling the class back as a whole group. If so, then determine after which questions this would be helpful.
- Consider if there are students who could benefit from working some of the questions in 13.9.2 Collaboration with you in a small group.

Literacy and Language Routines

Three Reads

Questions 3a and 3b in 13.9.2 Collaboration allows students to engage in a collaboration by taking turns reading the problem to each other, identifying key information to share with each other, and discussing their reasonings of classifications mathematically using the quadrilateral properties.

Mathematical Thinking and Reasoning (MTR) Standard

MA.K12.MTR.6.1: Reasonableness

Throughout the lesson, reasonableness is used in justifications of shapes using properties of quadrilaterals. In 13.9.2 question 3, as students decide what person to choose, they must be precise and explain their choice. Within length measurements in the unit, checking the calculations is important when determining if there are congruent measurements or not.

Differentiate Instruction

Allow students to use a graphing utility to help them when plotting points in determining classifications and properties of quadrilaterals. Students may need to have their properties list with them if they are lower-level learners who need that guidance to help with classifying the quadrilaterals. These visuals will assist students who benefit from visual entry points. Consider assigning partners for 13.9.2 questions 3a and 3b to have them engage in collaboration with one another on choosing the appropriate choice of who worked the problem correctly, and to justify classifications of the properties to the quadrilateral correctly. These discussions will help deepen the understanding of the lesson for auditory learners.

Lesson Notes

13.9.1 Warm-Up: Between Two Numbers

Component Overview: *Students will formulate their own values to substitute into the equation to make it reasonable.*

13.9.1 Warm-Up: Between Two Numbers

Sigsbee Deep	Britton Hill	Panorama Tower
Dwight Sigsbee led the first expedition to chart the floor of the Gulf of America. The Sigsbee Deep, the deepest region of the Gulf of America, was named after him. The exact maximum depth is not known, but is estimated to be between 3,750 meters (m) and 4,384 m.	Britton Hill, located in Walton County, is the highest point in the state of Florida. Britton Hill's elevation is 105 m.	Panorama Tower in Miami is the tallest building in Florida. It is 265 m tall.

Consider asking students if they have ever been to Britton Hill or the Panorama Tower.

1. Complete the inequality using the digits 0 – 9, without repeating, where $B =$ number of Britton Hills and $P =$ number of Panorama Towers.
   Answers will vary.

$$3750 < \boxed{0}\boxed{9}B + \boxed{1}\boxed{2}P < 4384$$

2. Use a different set of digits, without repeating.
   Answers will vary.

$$3750 < \boxed{1}\boxed{8}B + \boxed{0}\boxed{9}P < 4384$$

Consider extending this information into a discussion with the class to have students present their ideas.

- *What is your method of choosing the values?*
- *With which question was it easier for you to find the values?*

13.9.2 Collaboration: Real-World Quadrilaterals

Component Overview: *Students are asked to find lengths of segments in real-world situations, and classify the given quadrilateral using its properties.*

13.9.2 Collaboration: Real-World Quadrilaterals

1. Francesco is using computer software to design the layout of a house. The base of the house creates $DMHT$ with coordinates $D(-4,1)$, $M(4,5)$, $H(6,1)$, and $T(-2,-3)$.

   a. Complete the table. Show your work in the table. Leave answers in exact form.

Line Segment	Length
$\overline{DM}$	$\sqrt{(4+4)^2+(5-1)^2}$ $\sqrt{(8)^2+(4)^2}$ $\sqrt{64+16}$ $\sqrt{80}=4\sqrt{5}$
$\overline{MH}$	$\sqrt{(6-4)^2+(1-5)^2}$ $\sqrt{(2)^2+(-4)^2}$ $\sqrt{4+16}$ $\sqrt{20}=2\sqrt{5}$
$\overline{HT}$	$\sqrt{(-2-6)^2+(-3-1)^2}$ $\sqrt{(-8)^2+(-4)^2}$ $\sqrt{64+16}$ $\sqrt{80}=4\sqrt{5}$
$\overline{TD}$	$\sqrt{(-4+2)^2+(1+3)^2}$ $\sqrt{(-2)^2+(4)^2}$ $\sqrt{4+16}$ $\sqrt{20}=2\sqrt{5}$

This Collaboration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. Monitor while students are working to determine if there are any questions that need to be debriefed before moving to 13.9.3 Your Turn!. This could include highlighting student voices to share rich discussions or comments overheard while you circulated, or to address common misconceptions that were evident among the majority of the class. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed Collaboration.

Consider using the Instructional Routine, “Take Turns.” Each member of the group will take a turn at finding the length of the line segment. Together, the group will critique each other’s work, ask clarifying questions, and make corrections, if needed.

13.9.2 Collaboration: Real-World Quadrilaterals (continued)

b. Complete the table. Show your work in the table. Leave answers in exact form.

Diagonal Line Segment	Length
$\overline{DH}$	$\sqrt{(6+4)^2+(1-1)^2}$ $\sqrt{(10)^2+(0)^2}$ $\sqrt{100+0}$ $\sqrt{100}$ 10
$\overline{MT}$	$\sqrt{(-2-4)^2+(-3-5)^2}$ $\sqrt{(-6)^2+(-8)^2}$ $\sqrt{36+64}$ $\sqrt{100}$ 10

c. Finish the statement.
$DMHT$ is classified as a *rectangle* because ...

the diagonals are congruent with opposite sides being congruent and parallel.

d. If 1 unit on the coordinate plane equals 5 feet (ft), what are the dimensions of the base of the house? Round to the nearest thousandth.
$22.361\ ft \times 44.721\ ft$

2. Jillian is determining which type of tile to place on the floor. She wants a tile in the shape of a parallelogram. The vertices of the tile options are shown.

Tile A	Tile B
$(-11,6.5)$, $(-10,3.5)$, $(-6,6.5)$, and $(-7,9.5)$	$(3,9.5)$, $(7,6)$, $(9,8.5)$, and $(5,11)$

a. Which tile should Jillian pick if she wants a tile in the shape of a parallelogram?
Tile A

b. Explain your choice by listing the properties of parallelograms that were applied.
Answers may vary. I found the slope of the sides and determined that opposite sides are parallel for Tile A.

For students who finish early on question 1, challenge them to create a new house design, making sure their points make a house base. Students can swap with another peer and answer the same question as in question 1, and think about how the new problem relates to the original in the solutions of lengths or classifications.

Provide graph paper or a graphing utility for students on question 2, where they can graph the points for the tiles to determine which one will work and fit the description. Some students may also need a properties sheet, depending upon their fluency with it.

13.9.2 Collaboration: Real-World Quadrilaterals (continued)

3. Aharon and Shreya classified $RNPF$ with vertices at $R(-3,7)$, $N(-1,7)$, $P(-3,2)$, and $F(-5,4)$. Their classifications are shown.

Aharon's Classification	Shreya's Classification
$RNPF$ is a square or rectangle because the diagonals are congruent.	Quadrilateral $RNPF$ has congruent diagonals but cannot be classified as a parallelogram.

   a. Whose classification do you agree with? Justify your answer using properties.
   Shreya because more than just diagonals of a shape are needed to identify what type of quadrilateral it is.

   b. Write a note to the person whose classification is incorrect, explaining their mistake.
   Answers will vary.
   Side measurements of the shape should be considered to be able to identify the shape and not just the diagonals.

Consider guiding students through by asking the following questions:

- *What are the properties for a square and a rectangle?*
- *Do you need all of those properties to classify a quadrilateral as a square or rectangle? Why or why not?*

Consider providing struggling students with a table to keep their work organized and help them better see the connections between side lengths.

13.9.2 Collaboration: Real-World Quadrilaterals (continued)

4. $KVSW$ has vertices at $K(10,-4)$, $V(9,-8)$, $S(1,-4)$, and $W(6,-2)$.

   a. Show mathematically that $KVSW$ is a trapezoid, but not an isosceles trapezoid.

   Slope of $\overline{WK} = \frac{-2-(-4)}{6-10} = -\frac{2}{4} = -\frac{1}{2}$

   Slope of $\overline{SV} = \frac{-8-(-4)}{9-1} = -\frac{4}{8} = -\frac{1}{2}$

   Slope of $\overline{KV} = \frac{-4-(-8)}{10-9} = \frac{4}{1} = 4$

   Slope of $\overline{SW} = \frac{-4-(-2)}{1-6} = \frac{-2}{-5} = \frac{2}{5}$

   Quadrilateral $KVSW$ is a trapezoid since it has one set of parallel sides.

   $SW = \sqrt{(6-1)^2 + (-2+4)^2}$
   $SW = \sqrt{29}$

   $KW = \sqrt{(6-10)^2 + (-2+4)^2}$
   $KW = 2\sqrt{5}$

   $VK = \sqrt{(9-10)^2 + (-8+4)^2}$
   $VK = \sqrt{17}$

   $SV = \sqrt{(9-1)^2 + (-8+4)^2}$
   $SV = 4\sqrt{5}$

   Two sides are not congruent so the trapezoid is not an isosceles.

   b. List the properties of an isosceles trapezoid that did not apply to $KVSW$.

   The legs of an isosceles trapezoid are congruent.
   The diagonals of an isosceles trapezoid are congruent.

- *What property does Quadrilateral $KVSW$ have to make it a trapezoid?*
- *What would Quadrilateral $KVSW$ need in order for it to be an isosceles trapezoid?*

Consider providing students with graph paper or a graphing utility to provide as a visual.

13.9.2 Collaboration: Real-World Quadrilaterals (continued)

5. Ariyon was graphing square $MGNR$, but she mixed up her vertices. She knows that point M has coordinates $(3,5)$ and point G has coordinates $(7,7)$. Her two options for points N and R are shown.

Point N	Point R
(8,3)	(5,1)
(9,3)	(4,1)

a. Which coordinates are correct for points N and R? Justify your answer or show your work.
$N = (9,3)$ and $R = (5,1)$.

Slope of $\overline{MG}$ is $\frac{1}{2}$
Slope of $\overline{RN}$ is $\frac{1}{2}$
Slope of $\overline{GN}$ is -2
Slope of $\overline{MR}$ is -2

$$MG = \sqrt{(7-3)^2 + (7-5)^2} = 2\sqrt{5}$$

$$RN = \sqrt{(5-9)^2 + (1-3)^2} = 2\sqrt{5}$$

$$MR = \sqrt{(3-5)^2 + (5-1)^2} = 2\sqrt{5}$$

$$GN = \sqrt{(7-9)^2 + (7-3)^2} = 2\sqrt{5}$$

b. Which properties of a square did you use to determine the correct points N and R?
A square has opposite sides that are parallel and adjacent sides that are perpendicular and all four sides congruent.

- *What other properties can you use to determine the coordinates for points N and R?*
- *How did you determine which coordinates are correct for points N and R?*

13.9.3 Your Turn!

Component Overview: *Students are classifying quadrilaterals and find missing measurements of quadrilaterals using the vertices given.*

13.9.3 Your Turn!

1. $SCWK$ has vertices at $S(-3,-4)$, $C(0,-2)$, $W(0,-10)$, and $K(-3,-8)$.
   $SCWK$ is classified as a(n) _<u>isosceles trapezoid</u>___
   because ... *the bases are parallel and the two legs are congruent.*

2. Ziquan is designing a template for his 3D printer. He created two quadrilaterals using his software.

$TGHN$	$MWRB$
$T(16,6)$, $G(18,4)$, $H(17,-2)$, and $N(14,4)$	$M(4,5)$, $W(7,4)$, $R(8,-3)$, and $B(3,2)$

   a. Which quadrilateral can be classified as a kite? Show your work.
   Quadrilateral $MWRB$ is a kite.

   $MB = MW$
   $MW = \sqrt{(7-4)^2 + (4-5)^2} = \sqrt{10}$
   $MB = \sqrt{(3-4)^2 + (2-5)^2} = \sqrt{10}$

   $BR = WR$
   $BR = \sqrt{(3-8)^2 + (2+3)^2} = 5\sqrt{2}$
   $WR = \sqrt{(7-8)^2 + (4-(-3))^2} = 5\sqrt{2}$

   b. List the properties of a kite that you used.
   A kite has exactly two pairs of adjacent congruent sides.

   c. If 1 unit on the coordinate plane equals 6 inches (in), what are the measurements of the sides of the kite? Round to the nearest thousandth.
   $MB = 6(\sqrt{10}) = 18.974$
   $MW = 6(\sqrt{10}) = 18.974$
   $WR = 6(\sqrt{50}) = 42.426$
   $BR = 6(\sqrt{50}) = 42.426$

Question #1, students could put "trapezoid" instead of "isosceles trapezoid" because the bases are parallel, and the other two sides are not congruent, but this would not be precise.

Some students may need graph paper or a graphing utility. Also consider providing a table for students for their work.

Consider asking guiding questions for students who may be struggling:

- *What are the properties of a kite?*
- *What do you notice about the side lengths of the quadrilaterals?*

13.9.4 Wrap-Up: Reflection

Component Overview: *Students reflect on how they feel about what they have learned.*

13.9.4 Wrap-Up: Reflection

1. Place an **X** on the line in relation to how you feel about today's learning target.
   Answers will vary.

   - ☐ I can use coordinate geometry to classify quadrilaterals based on their properties in a mathematical or real-world context.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

Additional questions to assess students learning and understanding:

- *What is something that became clearer today about classifying quadrilaterals?*
- *What area of quadrilaterals within the lesson did you struggle with?*